Networks: LAN, WAN, internet, hardware and topologies

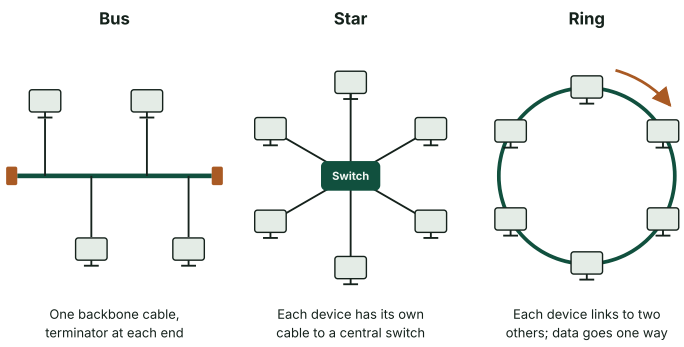
L16 FACT FILE

Specification: students should be able to

- Describe the main features of a local area network (LAN) and a wide area network (WAN)
- Describe the advantages and disadvantages of using a network in an organisation
- Describe the difference between the internet, the World Wide Web, an intranet and the Internet of Things
- Describe the function of a network interface card, network cables, a switch and a router
- Describe the bus, star and ring network topologies, including their advantages and disadvantages

CCEA fact file: Network Technologies 1, Network Technologies 2

Key words: network, standalone computer, LAN, WAN, file server, internet, World Wide Web, URL, intranet, Internet of Things, NIC, switch, router, IP address, topology, bus, terminator, star, ring



Bus, star and ring network topologies

Networks: the big picture

A network is a group of computers and devices linked by cable or wirelessly so they can share resources and communicate. A computer that is not linked to anything is called a standalone computer. A local area network (LAN) covers a small area, such as one school or office, and its hardware is usually owned by the organisation. A wide area network (WAN) covers a large geographical area and relies on telecommunications links such as fibre optic cables or satellites, often rented from other companies. The internet is the biggest WAN of all. Organisations use networks because users can log on anywhere, share printers and files, and have their data backed up and protected centrally. Networks also have costs: hardware must be bought, a technician is needed, and viruses can spread from one computer to all the others.

Internet, web, intranet and IoT

The internet and the World Wide Web are not the same thing. The internet is the global network of networks: the cables, routers and connections that carry data. The World Wide Web is one service that runs on the internet. It is made up of web pages, written in HTML and joined by hyperlinks, which are stored on web servers. Each site has a unique URL and pages are sent using HTTP or HTTPS. An intranet uses the same technology as the internet but is private. Only authorised users who log on with a username and password can use it, so it is useful for sharing staff documents, forms and messages. The Internet of Things (IoT) describes everyday objects, such as fitness watches, smart thermostats and video doorbells, that contain sensors and connect to the internet to send and receive data.

Network hardware

Every computer on a network needs a network interface card (NIC). This circuit board lets the computer connect to the network and gives it a port for a cable. Laptops and phones have a wireless NIC instead. Network cables physically link each NIC to another device. Most are copper twisted-pair cables in categories such as CAT5e and CAT6, where CAT6 can carry data faster. A switch is the central connection point in a LAN. It checks where each piece of data is going and forwards it only to the intended computer, using MAC addresses. A router connects separate networks, such as a school LAN and the internet. It reads the IP address on data and sends it along the best route to its destination. Many routers also include a firewall to filter traffic.

Topologies compared

A topology is the layout of a LAN. In a bus network every computer joins one backbone cable, with a terminator at each end to stop signals reflecting back. It is cheap and needs little cable, but if the backbone breaks the whole network stops, and collisions slow it down when it is busy. In a star network each computer has its own cable to a central switch (or server). A broken cable only affects one computer and it is easy to add more, but more cable is needed and the whole network fails if the central device fails. In a ring network each computer links to two neighbours to form a loop and data moves in one direction. There are no collisions, but one broken link stops the ring, and adding a computer means breaking the loop.

T Key terms from the mark schemes

LAN
Local area network: a network covering a small geographical area such as one building or site, usually owned by the organisation.
`Credited: Network in a small area, e.g. a school`

WAN
Wide area network: a network covering a large geographical area that uses telecommunications links such as fibre optic cables or satellites.
`Credited: Network over a large geographical area`

Standalone computer
A computer that is not connected to a network, so its files and devices cannot be shared.

Internet
The global network of networks: the hardware and connections that carry data between computers worldwide.
`Credited: A worldwide WAN`

World Wide Web
A collection of linked web pages stored on web servers and accessed over the internet using a browser.
`Credited: WWW`

Intranet
A private network owned by an organisation that uses internet technology but can only be accessed by authorised users who log on.
`Credited: Private network for an organisation's members`

Internet of Things (IoT)
Everyday physical devices with sensors and connectivity that send and receive data over the internet, e.g. fitness watches and smart thermostats.
`Credited: Everyday devices connected to the internet`

NIC
Network interface card: a circuit board in a computer that lets it connect to a network and provides a port for the network cable.
`Credited: Network adapter`

Switch
A device that connects computers on a LAN and forwards data only to the intended computer.
`Credited: Central connection point that sends data to the correct device`

Router
A device that connects networks, such as a LAN and the internet, and forwards data along the best route using IP addresses.
`Credited: Connects networks and directs packets`

Bus topology
A layout where all devices are attached to a single backbone cable with a terminator at each end.

Terminator
A device at each end of a bus backbone that absorbs the signal so it does not bounce back along the cable.

Star topology
A layout where each device is connected by its own cable to a central switch or server.

Ring topology
A layout where each device connects to two others in a closed loop and data travels in one direction.

Exam focus

This lesson appears on almost every Unit 1 paper. Topologies are the favourite: bus, star and ring diagrams or descriptions came up in 2019 Q6(b)–(d), 2022 Q7(a), 2023 Q7(c) and 2024 Q6(b). For a bus network, name the backbone and say what a terminator does; for a disadvantage, say the whole network fails if the backbone fails. Switch and router were asked in 2021 Q6 and 2023 Q7(b). Keep them separate: a switch forwards data to the right computer within a LAN, a router connects networks and uses IP addresses to route data. In 2025 Q8(a) many candidates could not expand IoT, so learn 'Internet of Things' and have an example ready. Internet versus WWW (2022 Q8(c), 2025 Q8(b)) needs both halves: network of networks versus pages accessed over it. 'Distinguish' questions need a comparison, not two separate definitions.

Unit 1 2025 Q13	[10]	Term matching: hardware, memory, networking, malware
Unit 1 2025 Q8	[10]	IoT, Internet vs WWW, uses of networks, networking concepts
Unit 1 2024 Q6	[13]	Network statements, bus disadvantages, LAN, encryption, HTTPS, RAM vs ROM, malware routes
Unit 1 2023 Q7	[8]	LAN, router and switch, bus diagram, NIC
Unit 1 2023 Q6	[5]	Communication technologies; IoT

? Check yourself

1. Southern Regional College links the computers in each of its campuses, and also links its Newry, Armagh and Banbridge campuses together. Describe two differences between the network inside one campus (a LAN) and the network joining the campuses (a WAN). [4]
2. Explain the difference between the internet and the World Wide Web. [2]
3. A grammar school in Coleraine has an intranet for its pupils and staff. Explain how the school's intranet differs from the internet. [4]

Answers: 1. Any two differences, each fully described (2 marks each): LAN covers a small geographical area such as one building/site (1) whereas a WAN covers a large geographical area such as countries/the world (1) / LAN hardware is usually owned by the organisation (1) whereas a WAN uses telecommunications links rented from other companies (1) / LAN uses network cables or Wi-Fi (1) whereas a WAN uses links such as fibre optic cables, satellites or telephone lines (1) 2. The internet is a global network of networks / the hardware and connections that carry data (1) / the WWW is a collection of web pages/websites that are accessed over the internet using a browser (1) 3. Internet: public / accessible to anyone with a communications link (1) / global network of networks (1) / Intranet: private network owned by an organisation (1) / only staff and pupils who log on with their own username and password can use it (1) / uses the same technology/protocols as the internet (1) / used to share internal information such as school resources and notices (1) – max 2 for each side, comparison needed for full marks